

DIPLOMATIC WORKING DOCUMENT — MODULAR ANNEX

ANNEX Z

In-Iran Transformation Pathways

Full Menu of Unilateral In-Country Disposition Options

Modular · Incorporable by Reference · No US Veto

Version 1.0 — June 2026

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Preamble

Purpose

The Memorandum of Understanding Pertaining to PEACE establishes in Part B-1.1 a menu of pathways for final disposition of Iran's 60% HEU stockpile. Among these, Pathways N, O, T, and W are in-country transformation pathways that Iran selects at any time as a unilateral, irrevocable right, without US concurrence, joint review, or veto (per MOU Article 3.2 and Article 1.9).

This Annex Z provides detailed technical specifications, timelines, verification protocols, and economic benefits for each in-Iran pathway. It also includes the Batch Splitting Provision allowing Iran to process the stockpile sequentially through multiple pathways.

All obligations execute in the active present tense. The IAEA performs verification. Step-back and earn-back follow Annex T. All technical pathways and financial drawdowns execute symmetrically and simultaneously with the Day-Zero Cryptographic Escrow established in Annex Y (v2.0).

Background

The parent framework — the MOU Pertaining to PEACE — establishes in Article 3.2 that the 60% stockpile neutralizes under verified custody on Iranian soil unless Iran chooses otherwise. The framework architecture respects the constraint that material does not leave Iran. The default disposition is in-country under the Dual-Key Custody Protocol (Annex Y). Iran holds an irrevocable, unilateral right to elect any In-Iran Transformation Pathway (N, O, T, or W) at any time, without US concurrence or veto.

The Joint Disposition Fund — a \$1 billion facility contributed equally by both Parties — rewards verified progress on disposition. Iran draws on the Fund as milestones verify, independent of which pathway it selects. No pathway is excluded from eligibility.

Relationship to Parent Framework

This Annex supplements and does not replace the disposition architecture of MOU Part B-1.1. Where this Annex provides specific technical protocols for in-Iran transformation pathways, the provisions of this Annex govern.

Modularity Clause

Modularity Clause

This Annex is self-contained and incorporates into any existing or future agreement between the Parties by a single sentence of adoption. It does not amend or supersede conflicting provisions of any parent framework unless the Parties explicitly so agree. In the event of inconsistency between this Annex and any parent framework provision, this Annex governs with respect to technical specifications, timelines, verification, and drawdown schedules for Pathways N, O, T, and W.

Article 1: Common Provisions

OPERATIVE AND BINDING

1.1 Unilateral Selection Right

Iran selects any of the pathways in this Annex Z by written notification to the PCC and the IAEA. The selection becomes operative 14 days after notification, subject only to IAEA technical clearance for safety, feasibility, and verification readiness. The United States has no veto, no consultation right beyond technical information, and no ability to delay or block the selection.

1.2 Material Accountancy Baseline

Before commencing any pathway, the IAEA completes a baseline inventory of the entire 60% HEU stockpile allocated to that pathway, using mass spectrometry and non-destructive assay. The baseline records in the Shared Digital Registry (Annex U).

1.3 Reversibility Assessment

The IAEA certifies for each pathway the point of no return — the stage after which the material cannot be reconstituted to weapons-usable form without extraordinary effort. After that point, the pathway is irreversible and counts fully toward Iran's disposition obligation.

1.4 Joint Disposition Fund Drawdown

Each pathway qualifies for drawdown from the \$1 billion Joint Disposition Fund (MOU Article 3.6). The schedule for each pathway is specified below: 50% on commencement, 50% on completion. Iran draws funds regardless of which pathway it selects, and no US approval is required beyond IAEA verification of milestones.

1.5 No US Custody, No US Soil

All operations occur in Iran under IAEA supervision. No US personnel exercise direct physical custody over material. US technical observers are present only with Iran's explicit, revocable, per-visit written consent (MOU Article 3.5).

1.6 Financial Liquidated Damages for Unilateral Withdrawal

In the event that the United States exercises a unilateral withdrawal from the parent framework in the absence of a certified Tier 2 or Tier 3 Iranian violation, or fails to maintain the structural sanctions waivers, all funds currently remaining in the \$1 Billion Joint Disposition Fund and the Annex W Maritime Escrow forfeit immediately and unconditionally to the sole sovereign control of the Islamic Republic of Iran without clawback, legal recourse, or requirement of IAEA technical clearance.

Article 2: Pathway N — Civilian Nuclear Products

OPERATIVE AND BINDING

2.1 Description

Iran downblends 60% HEU to low-enriched uranium ($\leq 5\%$ U-235) and fabricates it into civilian nuclear products, including:

- Reactor fuel assemblies for domestic use and export (for example: VVER fuel).
- Medical isotopes (molybdenum-99, iodine-131, etc.) via target irradiation.
- Betavoltaic batteries for pacemakers, satellites, and remote sensors.
- Research plates and foils for scientific applications.

2.2 Timeline

Phase	Duration	Activities
Preparation	3 months	Downblending equipment installation; IAEA seal verification; safety protocols
Downblending	9 months	Conversion of 60% HEU to LEU at Esfahan; continuous IAEA monitoring
Fabrication	6–18 months	Fuel assembly, isotope target, and battery production; product-specific lines
Total	18–30 months	Depending on product mix

2.3 Verification

IAEA continuous monitoring of downblending cascades, gamma spectroscopy on every batch, mass balance reconciliation monthly, and final product certification (fuel assembly serial numbers; isotope purity assays) confirm disposition progress throughout the pathway.

2.4 Reversibility

Low to negligible. Recovering weapons-usable material from fuel assemblies, isotopes, or betavoltaic batteries requires chemical reprocessing that the IAEA detects with near-100% probability. The point of no

return is the completion of downblending to $\leq 5\%$.

2.5 Economic Benefit

Iran retains all products for domestic use or export. Estimated annual revenue: \$20–\$50 million from fuel fabrication fees; \$5–\$20 million from isotope sales; additional revenue from batteries and research plates.

2.6 Joint Disposition Fund Drawdown

Milestone	Amount	Trigger
Commencement (IAEA certifies pathway start)	\$100M	IAEA technical clearance of pathway commencement
Completion (IAEA certifies first product batch)	\$100M	IAEA certification of irreversible transformation or point-of-no-return
Total	\$200M	

Article 3: Pathway O — Industrial Uranium Products

OPERATIVE AND BINDING

3.1 Description

Iran depletes 60% HEU to very low enriched uranium (0.3%–0.7% U-235) and processes it into industrial products, including:

- Rare earth element co-production feedstock: uranium leached simultaneously with rare earths (neodymium, dysprosium, terbium, europium) for electric vehicle motors and wind turbines.
- Uranium-carbon nanotube composites: for aerospace shielding, thermal management, and lightweight structural components.
- Petrochemical catalysts: uranium-based catalysts for petroleum refining, desulfurization, and ammonia production.

3.2 Timeline

Phase	Duration	Activities
Depletion	12 months	Downblending from 60% to 0.3–0.7% at Esfahan
Chemical processing	6–12 months	Leaching, co-production, composite synthesis
Product finishing	6–12 months	Formulation into marketable industrial materials
Total	24–36 months	Depending on product mix

3.3 Verification

The IAEA performs isotopic verification of depleted uranium ($\leq 0.7\%$), quarterly audits of industrial facilities (chain-of-custody), environmental sampling to confirm no diversion, and final product certification.

3.4 Reversibility

Negligible. Uranium is chemically bound in industrial matrices and depleted to levels far below natural uranium. Re-enrichment requires massive energy and chemical separation, detected easily by the IAEA.

3.5 Economic Benefit

Iran sells products to global industrial markets. Estimated revenue: rare earth oxides at \$50–\$500 per kilogram; composites at \$1,000–\$10,000 per kilogram; catalysts at \$5,000–\$50,000 per metric ton.

3.6 Joint Disposition Fund Drawdown

Milestone	Amount	Trigger
Commencement (IAEA certifies pathway start)	\$100M	IAEA technical clearance of pathway commencement
Completion (IAEA certifies first product batch)	\$100M	IAEA certification of irreversible transformation or point-of-no-return
Total	\$200M	

Article 4: Pathway T — Depleted Uranium Industrial

OPERATIVE AND BINDING

4.1 Description

Iran depletes 60% HEU to 0.3% U-235 and fabricates depleted uranium (DU) metal for industrial applications:

- Aircraft counterweights.
- Radiation shielding for medical and industrial facilities.
- Shipping container security (ballast and anti-penetration).
- Kinetic energy penetrators (non-nuclear military use, permitted as sovereign right).

4.2 Timeline

Phase	Duration	Activities
Depletion	12 months	Downblending to 0.3% U-235
Metal conversion	4 months	Reduction to DU metal via chemical processes
Total	16 months	

4.3 Verification

The IAEA performs isotopic verification ($\leq 0.3\%$), mass balance accounting, tamper-evident seals on stored metal, and annual physical inventory.

4.4 Reversibility

Very low to negligible. DU metal requires isotopic separation to recover U-235; the IAEA detects any enrichment attempt.

4.5 Economic Benefit

Iran sells DU metal to industrial buyers. Estimated market price: \$50–\$150 per kilogram.

4.6 Joint Disposition Fund Drawdown

Milestone	Amount	Trigger
Commencement (IAEA certifies pathway start)	\$50M	IAEA technical clearance of pathway commencement
Completion (IAEA certifies first product batch)	\$50M	IAEA certification of irreversible transformation or point-of-no-return
Total	\$100M	

Article 5: Pathway W — Agriculture and Food Security

OPERATIVE AND BINDING

5.1 Description

Iran depletes 60% HEU to natural-grade uranium (0.711% U-235) and combines it with phosphate rock to produce enhanced agricultural fertilizer. The uranium disperses across farmland, absorbs into crops, and enters the food supply. This is the most irreversible pathway.

The symbolism is direct: from bombs to bread. The same atoms that could have been weapons become wheat, rice, corn, and vegetables.

5.2 Timeline

Phase	Duration	Activities
Depletion	9 months	Downblending from 60% to 0.711% at Esfahan
Fertilizer production	4 months	Combining depleted uranium with phosphate rock; granulation
Field application	1 month	Dispersion across designated farmland under IAEA environmental monitoring
Total	14 months	Compressed with IAEA surge staffing

5.3 Verification

The IAEA performs isotopic verification (0.711% $\pm$ 0.05%), continuous monitoring of depletion and blending, sampling of final fertilizer batches, and environmental and agricultural safety protocols for soil application.

5.4 Reversibility

None. Recovering weapons-usable material would require collecting soil from millions of hectares, extracting trace uranium from a phosphate matrix (chemical and energy intensive), and re-enriching from

0.711% to 60% (thousands of centrifuge years). The IAEA declares the material permanently eliminated from proliferation risk at the point of field application.

5.5 Economic Benefit

Iran produces its own enhanced fertilizer, reducing import dependence and improving agricultural yields. Iran exports surplus fertilizer to regional markets.

5.6 Joint Disposition Fund Drawdown

Milestone	Amount	Trigger
Commencement (IAEA certifies pathway start)	\$35M	IAEA technical clearance of pathway commencement
Completion (IAEA certifies first product batch)	\$35M	IAEA certification of irreversible transformation or point-of-no-return
Total	\$70M	

5.7 Mandatory Destruction Floor (Phase I)

Regardless of Iran's pathway selection under MOU Article 3.2, a minimum of 10% of the 60% HEU stockpile (by mass) undergoes irreversible destruction via Pathway W. Execution occurs on Day 1 as an unseverable prerequisite for Phase I commencement.

The IAEA verifies the destruction of this mandatory floor using the standard Pathway W protocol: isotopic sampling of feed material, continuous monitoring of depletion and blending, and sampling of the final fertilizer. This mandatory 10% destruction floor constitutes a Day-Zero prerequisite. Upon signature of this Agreement, the Swiss-Omani Escrow Agent releases the \$100M from the Mutual Escrow Deposit into the humanitarian-restricted sub-account to serve only medicine, food, and medical equipment. This transaction executes via the Day-Zero Cryptographic Escrow simultaneously with the transfer of the Annex Y validation tokens, ensuring public relief lands on Day 1.

If Iran has selected Pathway W for the entire stockpile, this mandatory floor counts as satisfied upon the completion of the first 10% of that pathway. If Iran selects another pathway for the remaining stockpile, the 10% destruction floor remains separate and additional.

Failure to execute the Mandatory Destruction Floor on Day 1 triggers an automatic Tier 2 step-back under MOU Article 7.3, with a 10% reduction in non-humanitarian benefits for each additional 30-day period of non-completion, up to a maximum of 50%. The step-back reverses fully upon IAEA certification of completion.

5.8 Default Activation

Per MOU Article 3.2, if Iran has selected no final disposition pathway by Day 730, Pathway W activates automatically as the irreversible final backstop. No US veto blocks this activation.

Article 6: Batch Splitting

OPERATIVE AND BINDING

6.1 Right to Split

Iran splits the 60% HEU stockpile into up to three batches, processed sequentially (not in parallel). Each batch follows a different pathway from this Annex Z.

6.2 Notification

Iran notifies the PCC and the IAEA of the batch composition and sequencing at least 30 days before processing the first batch. PCC concurrence is not required — only notification. The IAEA verifies that splitting does not create verification gaps.

6.3 Sequential Processing

One batch completes substantially before the next batch commences. "Substantially complete" means the IAEA has certified the batch's point of no return (or full completion if irreversible).

6.4 Pathway Switching Mid-Batch

Iran switches a batch from one pathway to another with 30 days' notice to the PCC and IAEA. Funds already drawn from the Joint Disposition Fund under the original pathway credit against the new pathway's schedule with no clawback.

6.5 Material Accountancy

All batches remain under IAEA dual-key custody and material accountancy throughout. The IAEA tracks each batch separately.

Article 7: Relationship to Other Annexes

OPERATIVE AND BINDING

Annex	Relationship
MOU Part B-1.1	This Annex Z provides the technical detail for the pathways listed in the modular pathway menu.
Annex Y (Dual-Key Custody)	Material remains under dual-key custody until the point of no return for each pathway.
Annex T (Step-Back / Earn-Back)	Any violation in pathway execution triggers the Annex T tiered penalty system.
Annex U (Shared Digital Registry)	All milestones, certifications, and drawdowns are recorded in the Shared Digital Registry.
Annex V (Nuclear Dust)	This Annex Z applies to intact material; Annex V applies to entombed dust from damaged sites.
Annex Rc	Annex Rc governs the 60-day regional standstill along the Blue Line. Any certified violation of Annex Rc automatically pauses the countdown clocks for all remaining Joint Disposition Fund drawdowns under this Annex Z.

Article 8: Incorporation by Reference

OPERATIVE AND BINDING

This Annex Z incorporates into any existing or future agreement by the following adoption sentence:

"The Parties adopt Annex Z (In-Iran Transformation Pathways) as an operative provision of this Agreement. In the event of any inconsistency, Annex Z governs with respect to technical specifications, timelines, verification, and drawdown schedules for Pathways N, O, T, and W."

The Parties adopt Annex Y (v2.0) and Annex Z (v2.0), inclusive of the Day-Zero Cryptographic Escrow and the Annex Rc Supplementary Protocol, as operative, un-severable provisions of this Agreement. All technical pathways and financial drawdowns execute symmetrically and simultaneously. No amendment of the parent MOU is required. This Annex Z enters into force upon signature by both Parties or provisional application by exchange of diplomatic notes.

Signature Block

In witness whereof, the undersigned, being duly authorized by their respective governments, have signed this Annex Z: In-Iran Transformation Pathways.

Done at _____, this _____ day of _____, 2026, in triplicate in the English language.

For the Islamic Republic of Iran:

Name: _____

Title: _____

Date: _____

For the United States of America:

Name: _____

Title: _____

Date: _____

For the IAEA (as verification authority):

Name: _____

Title: _____

Date: _____

Witnessed by the Joint Secretariat (Muscat).

Annex Z

In-Iran Transformation Pathways

Full Menu of Unilateral In-Country Disposition Options

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